Lect. 8 Spatial Clustering Analysis

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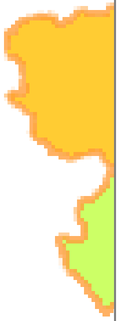

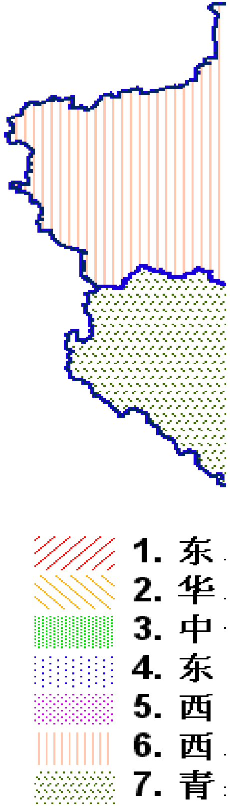

生态安全战略格局示意图

More cases of spatial clustering can be found on website :
https://mp.weixin.qq.com/s/0Lj-uSBBJWJqpuu_gFSxmw

Lect. 8 Spatial Clustering Analysis

- □ What is Clustering Analysis?
- □ How to Measure the Similarity of Clustering
- □ Spatial Clustering Approaches
    - Partitioning Clustering
    - Hierarchical Clustering
    - Density-based Clustering
    - Spatially Constrained Multivariate Clustering
- □ Cases

What is Clustering Analysis

- Cluster: a collection of data objects/features/points
  - Similar to one another within the same cluster
  - Dissimilar to the objects in other clusters
- Clustering analysis (or Clustering)
  - Discover similarities between data according to the characteristics found in the data and grouping similar data objects into clusters
- Two Types of Typical applications of Cluster Analysis
  - As a stand-alone tool to get insight into data distribution
  - As a preprocessing step for other algorithms

Clustering Analysis: Rich Applications and Multidisciplinary Efforts

- Pattern Recognition
- Spatial Data Analysis
  - Create thematic maps in GIS by clustering feature spaces
  - Detect spatial clusters or for other spatial mining tasks
- Image Processing e.g: identify areas of similar land use in an earth observation database
- Economic Science (especially market research)
- WWW
  - Document classification
  - Cluster Weblog data to discover groups of similar access patterns

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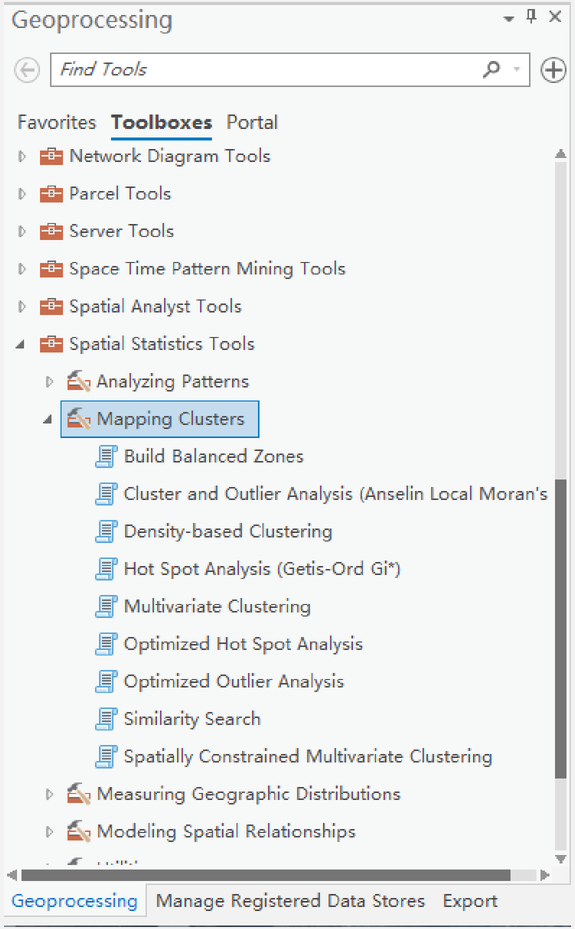
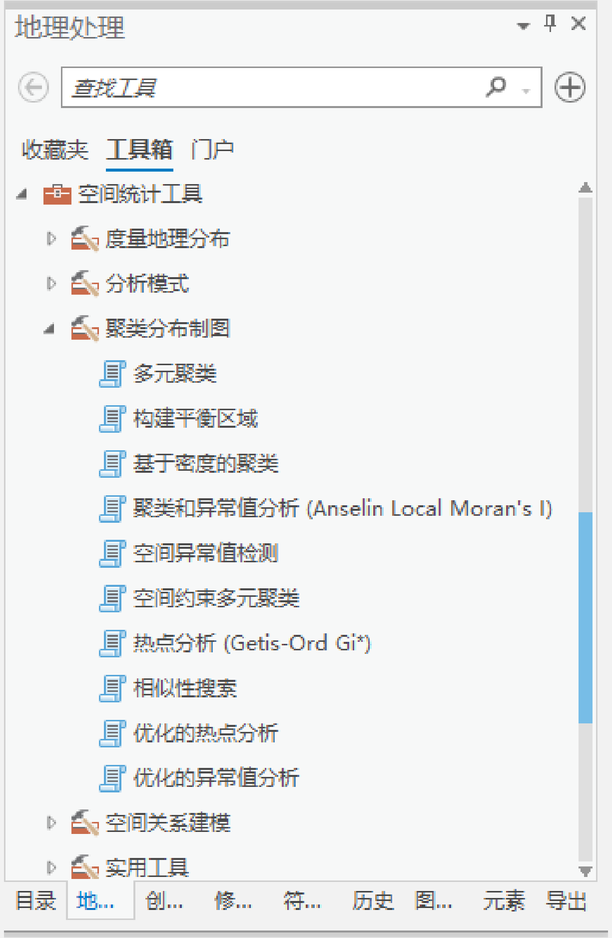
Pattern Recognition in ArcGIS Pro





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How to Measure the Similarity of Clustering

What is a good clustering method?

- A good clustering method will produce high quality clusters with
  - high intra-class similarity
  - low inter-class similarity
- The quality of a clustering result depends on both the similarity measure used by the method and its implementation.
- The quality of a clustering method is also measured by its ability to discover some or all of the hidden patterns.

How to Measure the Similarity of Clustering

Two types of methods for measuring similarity/dissimilarity of clustering: distance-based method and similarity-coefficient-based method

- **Distance-based method**
  - Similarity is expressed in terms of a **distance function**, typically metric: ***d*(*i*, *j*).**
  - The definitions of **distance functions** are usually very different for **interval-scaled,** boolean, categorical, ordinal, ratio, and vector variables.
  - Weights should be associated with different variables based on applications and data semantics.
  - It is hard to define "similar enough" or "good enough".
    - The answer is typically highly subjective.

Distance-based method

Some popular distances for measuring the (dis)similarity between groups:

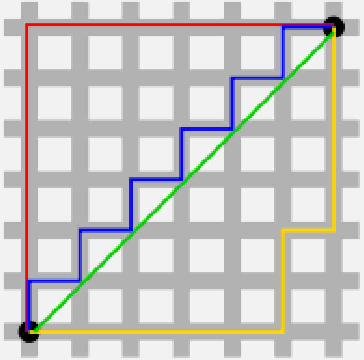
- **Minkowski distance** :

$$d(i,j)=\sqrt[q]{(|x_{i_1}-x_{j_1}|^q+|x_{i_2}-x_{j_2}|^q+\ldots+|x_{i_p}-x_{j_p}|^q)}$$

where $(x_{i1}, x_{i2}, \ldots, x_{ip})$ and $(x_{j1}, x_{j2}, \ldots, x_{jp})$ are two p-dimensional data objects, and q is a positive integer

- **Manhattan distance** ($q = 1$; L1 distance, 1-norm of vector)
  - Aka. city block distance:

$$d(i,j)=|x_{i_1}-x_{j_1}|+|x_{i_2}-x_{j_2}|+\ldots+|x_{i_p}-x_{j_p}|$$



Distance-based method

Some popular distances for measuring the (dis)similarity between groups:

- **Euclidean distance** (*q = 2*; L2 distance, 2-norm of vector)

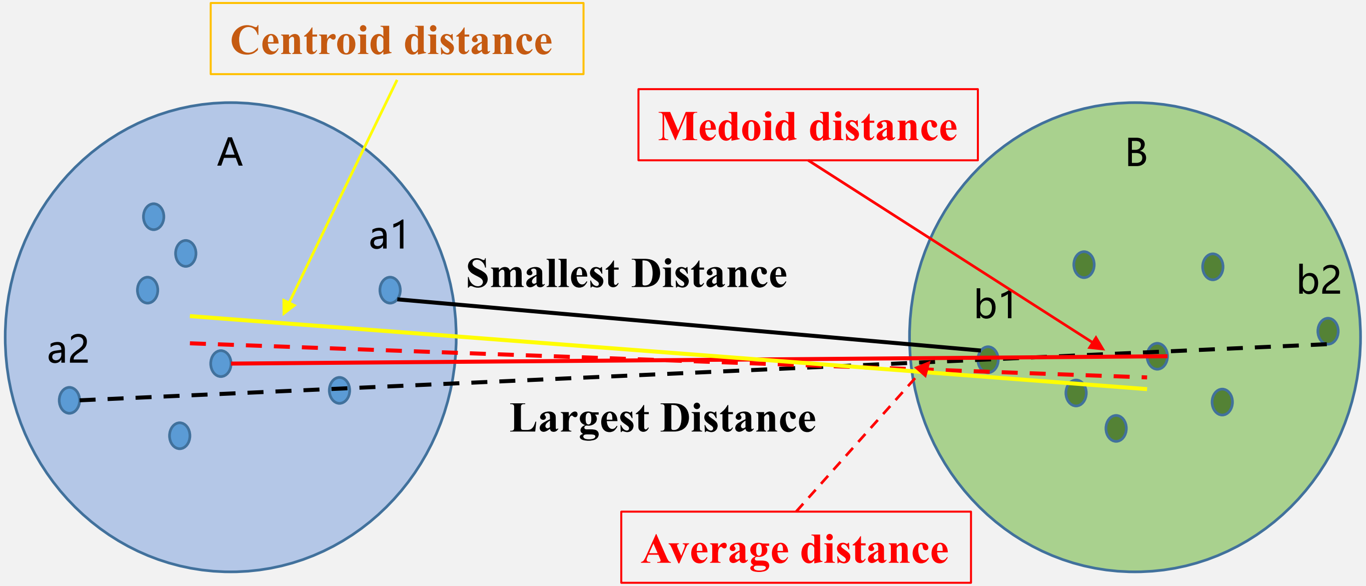
$$d(i,j)=\sqrt{(|x_{i_1}-x_{j_1}|^2+|x_{i_2}-x_{j_2}|^2+\ldots+|x_{i_p}-x_{j_p}|^2)}$$

- Properties
  - $d(i,j) \geq 0$
  - $d(i,i) = 0$
  - $d(i,j) = d(j,i)$
  - $d(i,j) \leq d(i,k) + d(k,j)$

Typical Alternatives to Calculate the Distance between Clusters

- **Smallest distance:** smallest distance between an element in one cluster and an element in the other, i.e., $dis(K_i, K_j) = \min(t_{ip}, t_{jq})$
- **Largest distance:** largest distance between an element in one cluster and an element in the other, i.e., $dis(K_i, K_j) = \max(t_{ip}, t_{jq})$
- **Average distance :** average distance between an element in one cluster and an element in the other, i.e., $dis(K_i, K_j) = avg(t_{ip}, t_{jq})$
- **Centroid distance :** distance between the centroids of two clusters, i.e., $dis(K_i, K_j) = dis(C_i, C_j)$. Centroid is the "middle" of a cluster
- **Medoid distance :** distance between the medoids of two clusters, i.e., $dis(K_i, K_j) = dis(M_i, M_j)$
  - Medoid: one chosen, centrally located object in the cluster
- ….

Typical Alternatives to Calculate the Distance between Clusters： Illustration

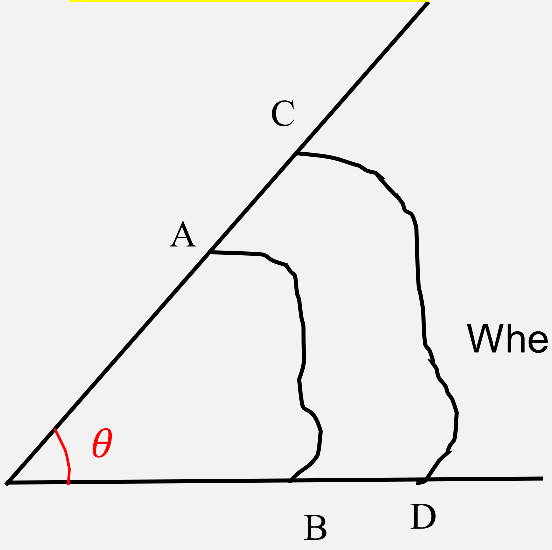


Similarity-Coefficient-Based Method

Angle cosine (夹角余弦) and correlation coefficient ***r*** are two classes of commonly used similarity coefficients.

Angle Cosine

Suppose: The numbers of objects and variables are ***m*** and ***n***, respectively.



$$\cos\theta_{ij} = \frac{\sum_{k=1}^{n} x_{ik} x_{jk}}{\sqrt{\sum_{k=1}^{n} x_{ik} \bullet \sum_{k=1}^{n} x_{jk}}}$$

Where *i* and *j* are ordinal numbers of two objects, x_k represents the k^{th} *variable.*

$$\theta = \begin{pmatrix} \cos\theta_{11} & \cos\theta_{12} & \cdots & \cos\theta_{1m} \\ \cos\theta_{21} & \cos\theta_{22} & \cdots & \cos\theta_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ \cos\theta_{m1} & \cos\theta_{m2} & \cdots & \cos\theta_{mm} \end{pmatrix}$$

$\cos\theta_{ij} \in [-1, 1]$ the larger the values of angle cosine, the more similar the objects.

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Main Types of Clustering Approaches--1

Clustering approaches include partitioning approach, hierarchical approach, density-based approach, and constraint-based approach, etc.

1. Partitioning approach:

- Constructs various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
- Typical methods: ***k*-means**, ***k*-medoids, ISODATA**, CLARANS

2. Hierarchical approach:

- Create a hierarchical decomposition of the set of data (or objects) using some criterion
- Typical methods: **AGNES, DIANA**, BIRCH, ROCK, CAMELEON

3. Density-based approach

- Based on density functions
- Typical methods: **DBSACN**, **OPTICS**, **HDBSCAN,**

Main Types of Clustering Approaches--2

4. Constraint-based Approach:

- Cluster objects by considering user-specified or application-specific constraints
- Typical methods: **Spatially Constrained Clustering,** COD (obstacles)

5. Model-based Approach:

- A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
- Typical methods: SOM (Self Organized Maps), EM, COBWEB

6. Raster-based or vector-based Approach

- **Raster-based**
- **Vector-based**

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Partitioning Clustering

- Construct a partition of a database ***D*** of ***n*** objects into a set of ***k*** clusters, s.t., minimum sum of squared distance

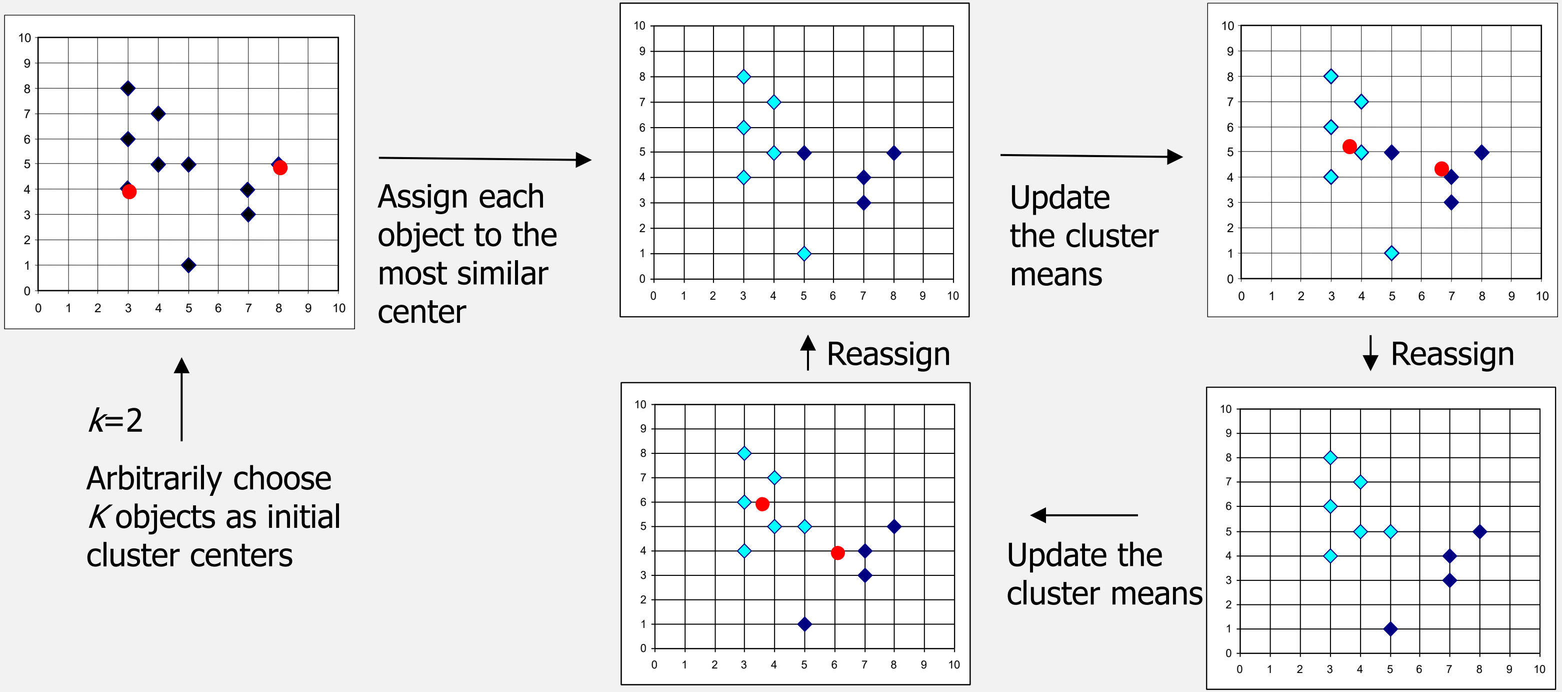
$$\text{s.t. } \sum_{m=1}^{k} \sum_{i=1}^{n} (C_m - v_{mi})^2$$

C_m is the middle of cluster m,
v_{mi} is the ith object value in cluster m.

- Given a *k*, find a partition of *k clusters* that optimizes the chosen partitioning criterion
  - Global optimal: exhaustively enumerate all partitions
  - Heuristic methods: ***k*-means** and ***k*-medoids** algorithms
    - ***k*-means** (*MacQueen'67*): Each cluster is represented by the center of the cluster
    - ***k*-medoids** / **PAM** (Partition Around Medoids) (*Kaufman & Rousseeuw'87*): Each cluster is represented by one of the objects in the cluster
    - **ISODATA**

The *K*-Means Clustering Method

- **Illustration**



The *K*-Means Clustering Method

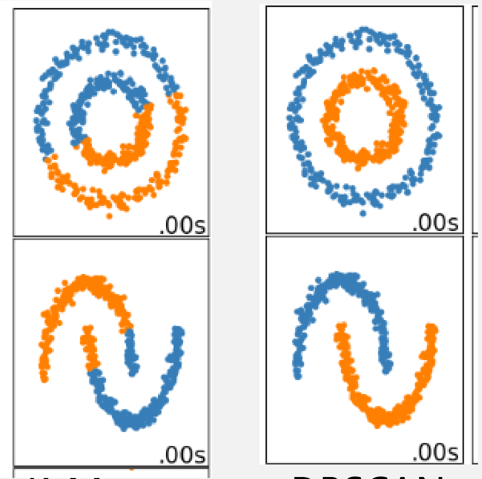
- Given a ***k***, the ***k-means* algorithm** is implemented in four steps:

  1. Arbitrarily choose *k* objects as initial *k* cluster centers
  2. Assign each object to the most similar center;
  3. Update the centroids of the clusters of the current partition (the centroid is the center, i.e., *mean point*, of the cluster);
  4. Go back to step 2, stop when no more new assignment.

Comments on The *K*-Means Clustering Method

- **Strength:** *Relatively efficient*: $O(tkn)$, where n is # objects, k is # clusters, and t is # iterations. Normally, k, $t << n$.
- Comment: Often terminates at a *local optimum*.
- **Weakness:**
  - **Unable to handle noisy data and *outliers***
  - Applicable only when mean is defined, then what about categorical data?
  - Need to specify *k, the number* of clusters, in advance
  - Not suitable to discover clusters with *non-convex shapes*

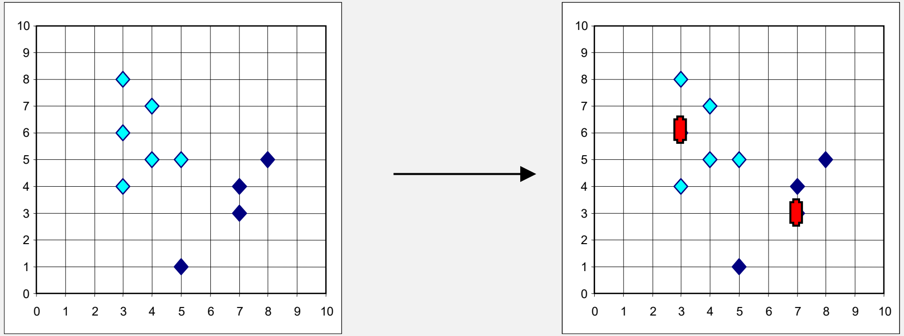
Non-convex shape data



K-Means DBSCAN

The *K*-Medoids Clustering Method

- The *k*-means algorithm is sensitive to outliers.
  - Since an object with an extremely large value may substantially distort the distribution of the data.
- ***K*-Medoids**: Instead of taking the **mean** value of the object in a cluster as a reference point, **medoid** can be used, which is the **most centrally located** object in a cluster.

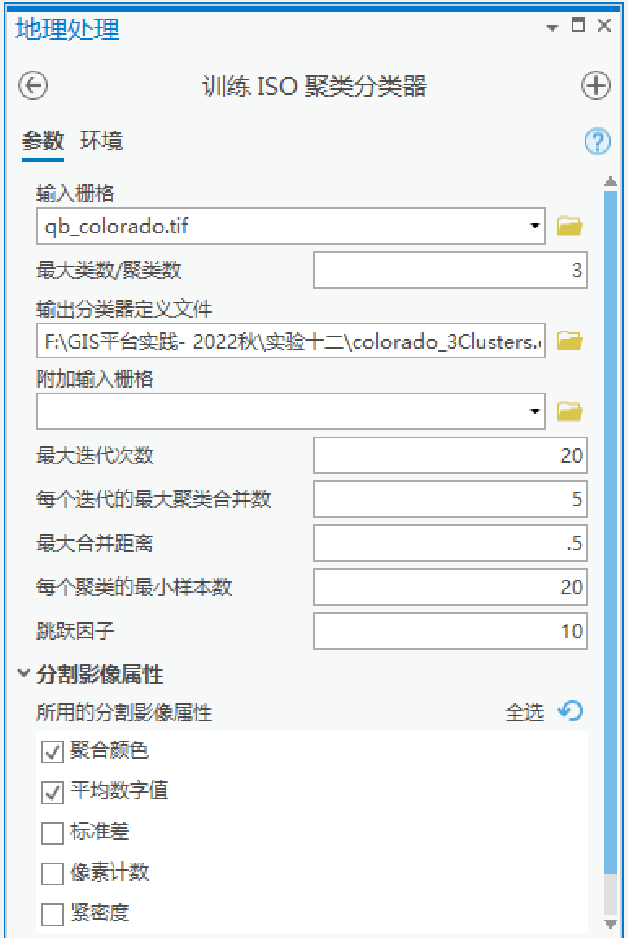


ISOData: Iterative Selforganizing Data Analysis Techniques Algorithm

ISODATA is based on *K*-Means algorithm, it adds two operations (“merging” and “splitting”) to the clustering results, and sets the algorithm to run the control parameters.

- **Merging operation:**
  - When the distance between two certain clusters (centers /classes) is less than a given threshold, these two clusters/classes are merged into one.
  - The number of samples of a certain cluster/class exceeds a given threshold, this cluster/class is merged into the neighboring cluster/class .
- **Splitting operation:**
  - When the standard deviation of a certain cluster/class is greater than a given threshold, or the number of its samples exceeds a given threshold, this cluster/class is divided into two.

ISO (ISOData) method in ArcGIS Pro



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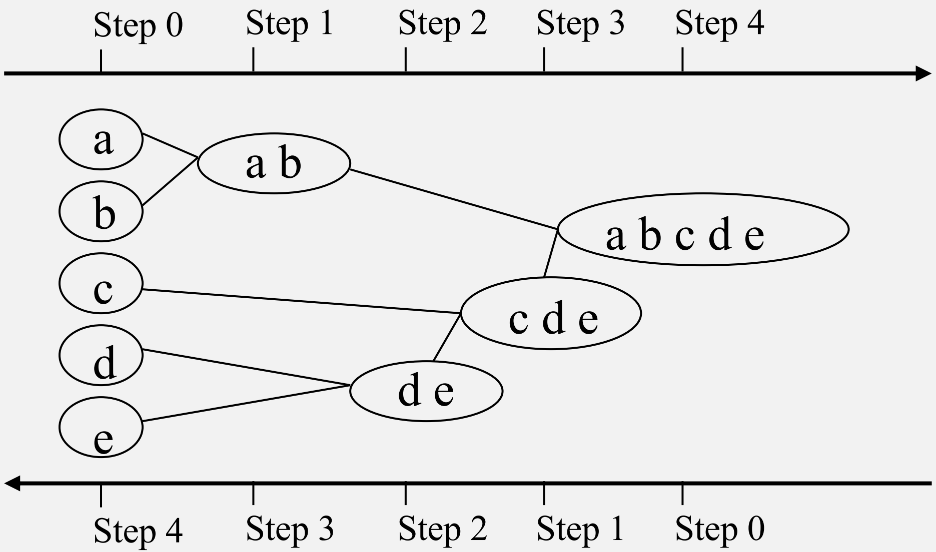
Hierarchical Clustering

- Use distance matrix as clustering criteria.
- This method does not require the number of clusters *k* as an input, but needs a termination condition

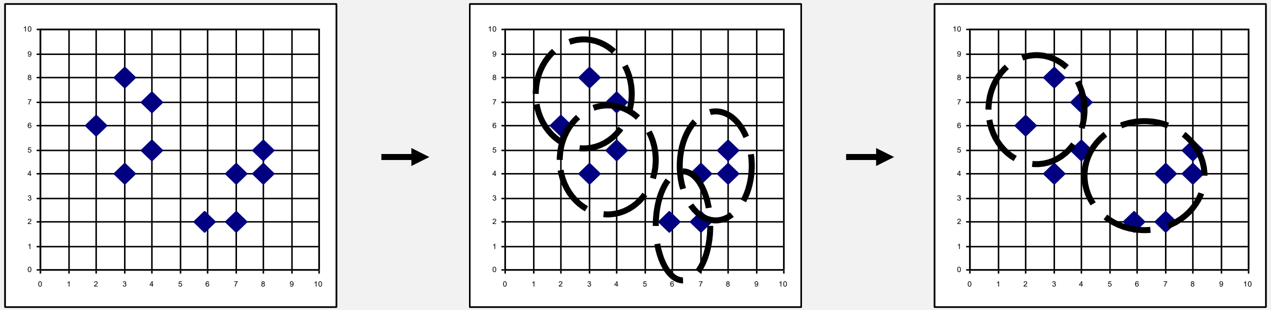


AGglomerative NESting (AGNES)
Bottom-up
聚合层次分类

DIvisive ANAlysis (DIANA)
Top-down
分裂层次分类

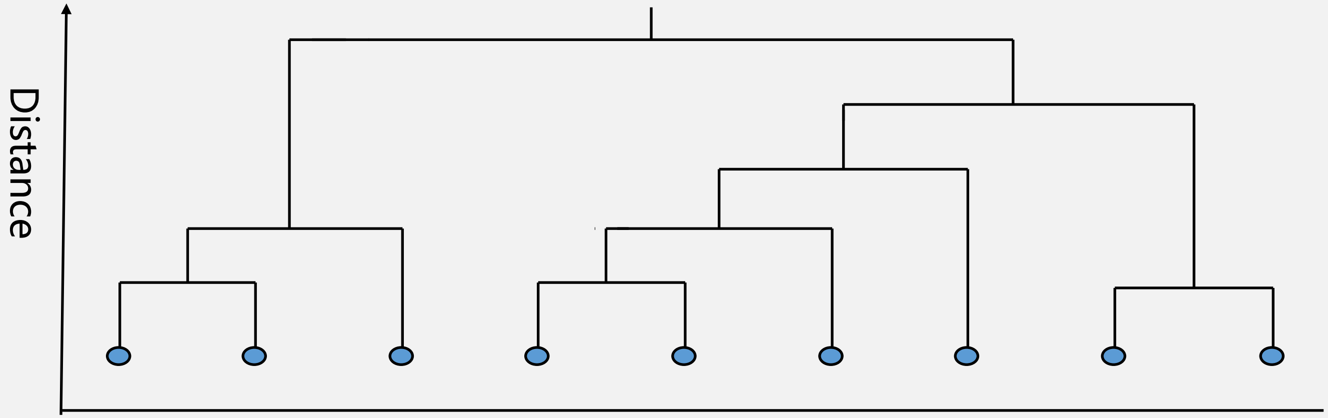
AGNES (AGglomerative NESting)

- Implemented in statistical analysis packages, e.g., SPSS, Splus
- Processing:
  1) View each object/ node as a cluster;
  2) Use the smallest-distance method and dissimilarity matrix to merge nodes (objects) that have the least dissimilarity;
  3) Go on in a non-descending fashion;
  4) Eventually all nodes belong to the same cluster.



Dendrogram: Shows How the Clusters are Merged in AGNES

- Decompose data objects into a several levels of nested partitioning (tree of clusters), called a dendrogram.
- A clustering of the data objects is obtained by cutting the dendrogram at the desired level, then each connected component forms a cluster.



Dendrogram

Example of AGNES

Suppose we have the dissimilarity matrix for 9 agricultural areas, use AGNES (AGglomerative NESting) method to hierarchically cluster the 9 agricultural areas into 2 groups.

$$D=(d_{ij})_{9\times 9}=\begin{bmatrix} 0 & & & & & & & & \\ 1.52 & 0 & & & & & & & \\ 3.10 & 2.70 & 0 & & & & & & \\ 2.19 & 1.47 & 1.23 & 0 & & & & & \\ 5.86 & 6.02 & 3.64 & 4.77 & 0 & & & & \\ 4.72 & 4.46 & 1.86 & 2.99 & 1.78 & 0 & & & \\ 5.79 & 5.53 & 2.93 & 4.06 & 0.83 & 1.07 & 0 & & \\ 1.32 & 0.88 & 2.24 & 1.29 & 5.14 & 3.96 & 5.03 & 0 & \\ 2.62 & 1.66 & 1.20 & 0.51 & 4.84 & 3.06 & 3.32 & 1.40 & 0 \end{bmatrix}$$

Example of AGNES

Solve:

Step 1, since d49=d94=0.51 is the smallest among all elements, merge areas 4 and 9 into a cluster and rule out row 9

Step2, since d75= d57=0.83 is the smallest among all rest elements, merge areas 5 and 7 into a cluster and rule out row 7

Step3, since d82= d28=0.88 is the smallest among all rest elements, merge areas2 and 8 into a cluster and rule out row 8

Step4, since d43= d34=1.23 is the smallest among all rest elements, merge areas3 and 4 into a cluster and rule out row 4. Now areas 3,4 and 9 are attributed into a same cluster.

Step5, since d21= d12=1.52 is the smallest among all rest elements, merge areas1 and 2 into a cluster and rule out row 2. Now areas 1,2 and 8 are attributed into a same cluster.

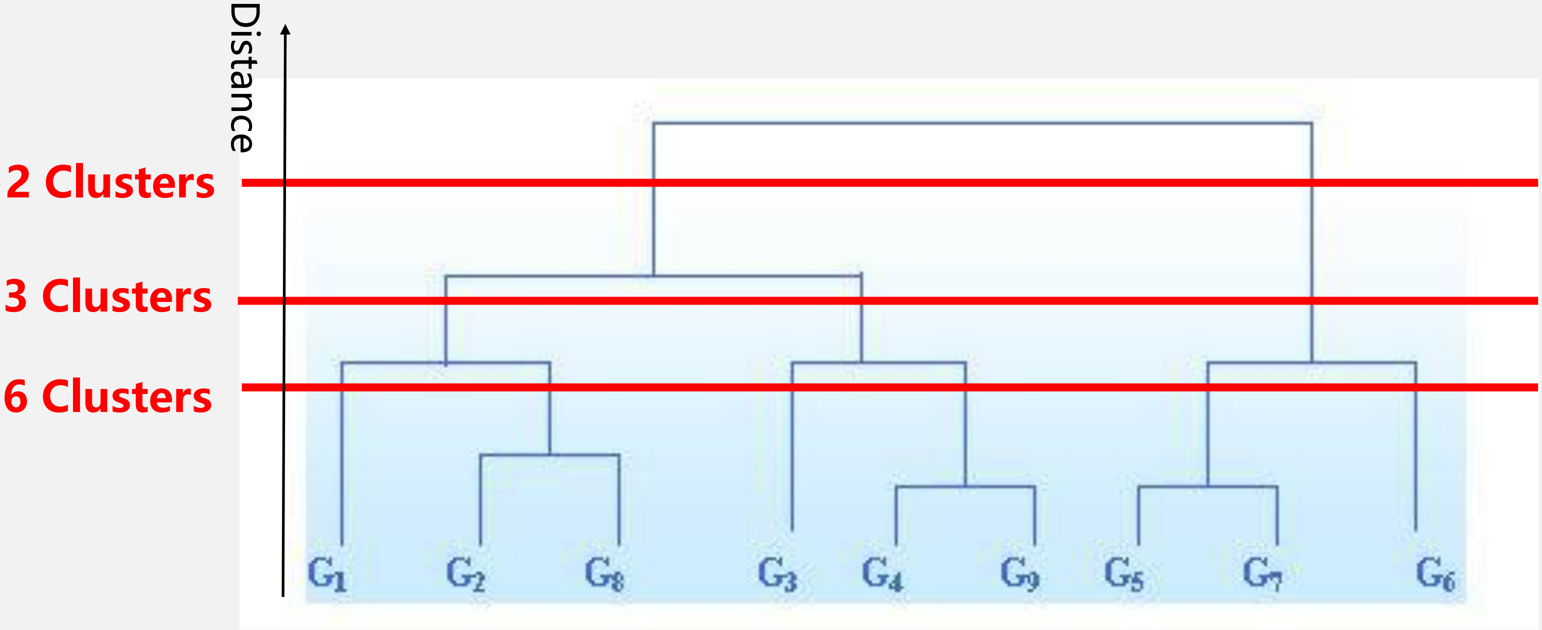
Step6, since d65= d56=1.78 is the smallest among all rest elements, merge areas 5 and 6 into a cluster and rule out row 6.Now areas 5,6 and 7 are attributed into a same cluster.

Step7, since d32= d23=2.70 is the smallest among all rest elements, merge areas 2 and 3 into a cluster and rule out row 3.Now areas 2, 3,4,8,9 are attributed into a same cluster.

Step8, since d53= d35=3.68 is the last node which has not been ruled out, merge areas 3 and 5 into a cluster and rule out row 5. Now areas 1,2,3,4,5, 6,7,8,9 are attributed into a same cluster.

$$D=(d_{ij})_{9\times 9}=\begin{bmatrix} 0 \\ 1.52 & 0 \\ 3.10 & 2.70 & 0 \\ 2.19 & 1.47 & 1.23 & 0 \\ 5.86 & 6.02 & 3.64 & 4.77 & 0 \\ 4.72 & 4.46 & 1.86 & 2.99 & 1.78 & 0 \\ 5.79 & 5.53 & 2.93 & 4.06 & 0.83 & 1.07 & 0 \\ 1.32 & 0.88 & 2.24 & 1.29 & 5.14 & 3.96 & 5.03 & 0 \\ 2.62 & 1.66 & 1.20 & 0.51 & 4.84 & 3.06 & 3.32 & 1.40 & 0 \end{bmatrix}$$

Example of AGNES



2 Clusters schema:

Cluster 1: 1,2, 3,4, 8,9

Cluster 2: 5,6,7

Modified AGNES (AGglomerative NESting)

1） View each node in a *m*×*m dissimilarity matrix* as a cluster;

2） Use the smallest-distance method and dissimilarity matrix to merge clusters/nodes that have the least dissimilarity into a new cluster; $G_k=\{G_p, G_q)$

$$d_{pq} = \min\{d_{ij}\}$$

3） Calculate the dissimilarity distances between the new cluster and the other nodes/clusters respectively to get a (*m*-1) ×(*m-1*) matrix；

$$d_{rk} = \min\{d_{rp}, d_{rq}\}\ (k \neq p, q)$$

$$\text{or } d_{rk} = \max\{d_{rp}, d_{rq}\}\ (k \neq p, q)$$

4) Go on in a non-descending fashion;

5） Eventually all nodes belong to the same cluster.

Example of Modified AGNES

Solve：

$$D=(d_{ij})_{9\times 9}=\begin{bmatrix} 0 &&&&&&&& \\ 1.52 & 0 &&&&&&& \\ 3.10 & 2.70 & 0 &&&&&& \\ 2.19 & 1.47 & 1.23 & 0 &&&&& \\ 5.86 & 6.02 & 3.64 & 4.77 & 0 &&&& \\ 4.72 & 4.46 & 1.86 & 2.99 & 1.78 & 0 &&& \\ 5.79 & 5.53 & 2.93 & 4.06 & 0.83 & 1.07 & 0 && \\ 1.32 & 0.88 & 2.24 & 1.29 & 5.14 & 3.96 & 5.03 & 0 & \\ 2.62 & 1.66 & 1.20 & \boxed{0.51} & 4.84 & 3.06 & 3.32 & 1.40 & 0 \end{bmatrix}$$

Step1:

- Since d49=d94=0.51 is the smallest, merge G_4 and G_9 into new cluster G_{10}, $G_{10}=\{G_4, G_9\}$.
- Calculate the dissimilarity distances between cluster G_{10} and the other clusters (1-3, 5-8) respectively to get an 8×8 matrix：

$$d_{1,10}=\min\{d_{14}, d_{19}\}=\min\{2.19, 2.62\}=2.19$$

$$d_{2,10}=\min\{d_{24}, d_{29}\}=\min\{1.47, 1.66\}=1.47$$

$$d_{3,10}=\min\{d_{34}, d_{39}\}=\min\{1.23, 1.20\}=1.20$$

$$d_{5,10}=\min\{d_{54}, d_{59}\}=\min\{4.77, 4.84\}=4.77$$

$$d_{6,10}=\min\{d_{64}, d_{69}\}=\min\{2.99, 3.06\}=2.99$$

$$d_{7,10}=\min\{d_{74}, d_{79}\}=\min\{4.06, 3.32\}=3.32$$

$$d_{8,10}=\min\{d_{84}, d_{89}\}=\min\{1.29, 1.40\}=1.29$$

Example of Modified AGNES

Step 2:

- Since d57=D75= 0.83 is the smallest, merge G_5 and G_7 into new cluster G_{11}, G_{11}= $\{G_5, G_7\}$.
- Calculate the dissimilarity distances between cluster G_{11} and the other clusters respectively to get a 7×7 matrix.

	G_1	G_2	G_3	G_5	G_6	G_7	G_8	G_{10}
G_1	0							
G_2	1.52	0						
G_3	3.10	2.70	0					
G_5	5.86	6.02	3.64	0				
G_6	4.72	4.46	1.86	1.78	0			
G_7	5.79	5.53	2.93	$\boxed{0.83}$	1.07	0		
G_8	1.32	0.88	2.24	5.14	3.96	5.03	0	
G_{10}	2.19	1.47	1.20	4.77	2.99	3.32	1.29	0

	G_1	G_2	G_3	G_6	G_8	G_{10}	G_{11}
G_1	0						
G_2	1.52	0					
G_3	3.10	2.70	0				
G_6	4.72	4.46	1.86	0			
G_8	1.32	$\boxed{0.88}$	2.24	3.96	0		
G_{10}	2.19	1.47	1.20	2.99	1.29	0	
G_{11}	5.79	5.53	2.93	1.07	5.03	3.32	0

Example of Modified AGNES

Step 3:

- Since d2,8=0.88 is the smallest, merge G_2 and G_8 into new cluster G_{12}, $G_{12}=\{G_2, G_8\}$.
- Calculate the dissimilarity distances between cluster G_{12} and the other clusters respectively to get a 6×6 matrix

	G_1	G_3	G_6	G_{10}	G_{11}	G_{12}
G_1	0					
G_3	3.10	0				
G_6	4.72	1.86	0			
G_{10}	2.19	1.20	2.99	0		
G_{11}	5.79	2.93	$\boxed{1.07}$	3.32	0	
G_{12}	1.32	2.24	3.96	1.29	5.03	0

Step 4:

- Since d6,11=1.07 is the smallest, merge G_6 and G_{11} into new cluster G_{13}, $G_{13}=\{G_6, G_{11}\}$.
- Calculate the dissimilarity distances between cluster G_{13} and the other clusters respectively to get a 5×5 matrix

	G_1	G_3	G_{10}	G_{12}	G_{13}
G_1	0				
G_3	3.10	0			
G_{10}	2.19	$\boxed{1.20}$	0		
G_{12}	1.32	2.24	1.29	0	
G_{13}	4.72	1.86	2.99	3.96	0

Example of Modified AGNES

Step 5:

- Since d3,10=1.20 is the smallest, merge G_3 and G_{10} into new cluster G_{14}, G_{14}= {G_3, G_{10}} .
- Calculate the dissimilarity distances between cluster G_{14} and the other clusters respectively to get a 4×4 matrix

	G_1	G_{12}	G_{13}	G_{14}
G_1	0			
G_{12}	1.32	0		
G_{13}	4.72	3.96	0	
G_{14}	2.19	$\boxed{1.29}$	1.86	0

Step 6:

- Since d12,14=1.29 is the smallest, merge G_{12} and G_{14} into new cluster G_{15}, G_{15}= {G_{12}, G_{14}} .
- Calculate the dissimilarity distances between cluster $G_{15}$and the other clusters respectively to get a 3×3 matrix

	G_1	G_{13}	G_{15}
G_1	0		
G_{13}	4.72	0	
G_{15}	$\boxed{1.32}$	1.86	0

Example of Modified AGNES

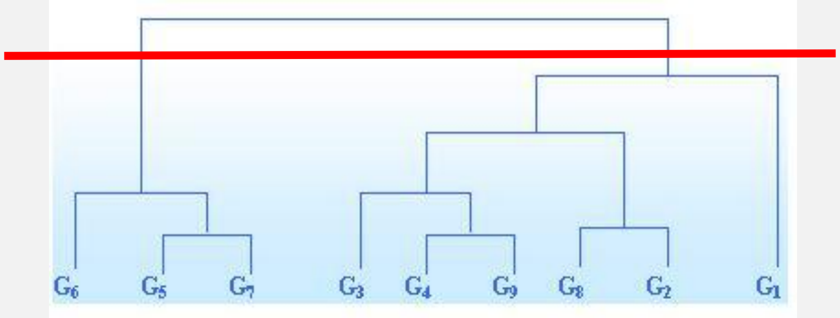
Step 7:

- Since d1,15=1.32 is the smallest, merge G_1 and G_{15} into new cluster G_{16}, G_{16}= {G_1, G_{15}} .
- Calculate the dissimilarity distances between cluster G_{16} and the other clusters respectively to get a 2×2 matrix

	G_{13}	G_{16}
G_{13}	0	
G_{16}	1.32	0

Step 8:

- Since there are only two clusters (G_{13} and G_{16}) remaining, merger they into new cluster G_{17}, G_{17} = {G_{13}, G_{16}} .
- Draw the dendrogram.



2 Clusters schema:

Cluster 1: 1,2, 3,4, 8,9

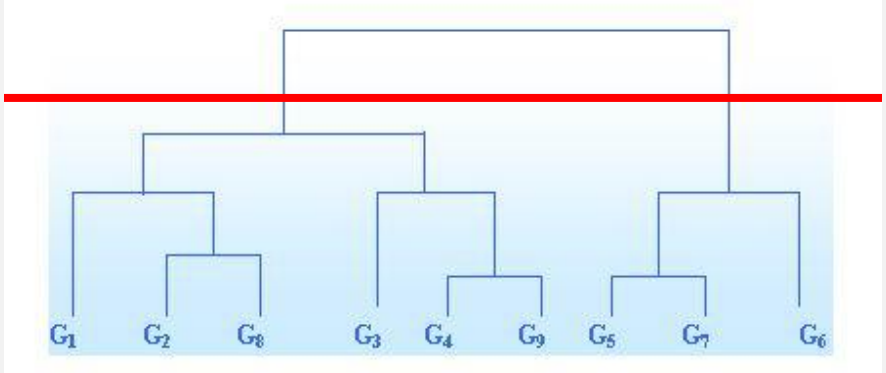
Cluster 2: 5,6,7

Example of Modified AGNES

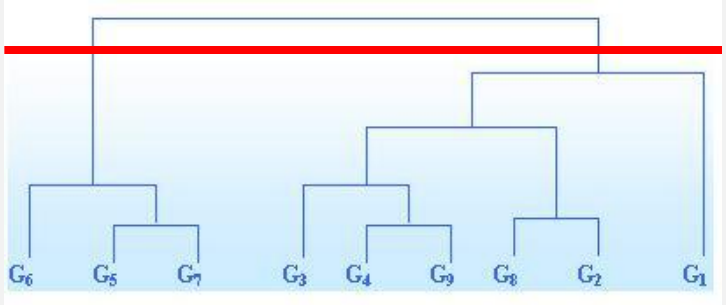
How about using maximum-distance to cluster 9 agricultural areas？

$$D=(d_{ij})_{9\times 9}=\begin{bmatrix} 0 \\ 1.52 & 0 \\ 3.10 & 2.70 & 0 \\ 2.19 & 1.47 & 1.23 & 0 \\ 5.86 & 6.02 & 3.64 & 4.77 & 0 \\ 4.72 & 4.46 & 1.86 & 2.99 & 1.78 & 0 \\ 5.79 & 5.53 & 2.93 & 4.06 & 0.83 & 1.07 & 0 \\ 1.32 & 0.88 & 2.24 & 1.29 & 5.14 & 3.96 & 5.03 & 0 \\ 2.62 & 1.66 & 1.20 & 0.51 & 4.84 & 3.06 & 3.32 & 1.40 & 0 \end{bmatrix}$$

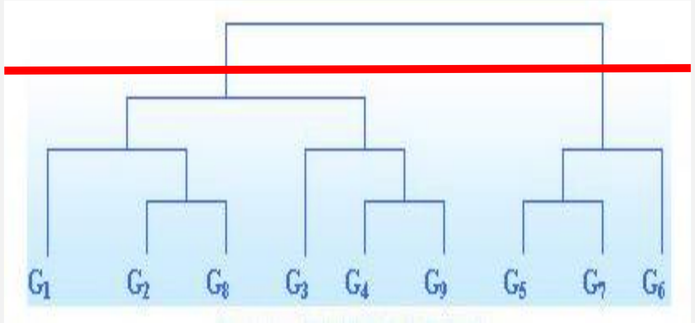
Comparison of the results from AGNES and its modified method



Original method

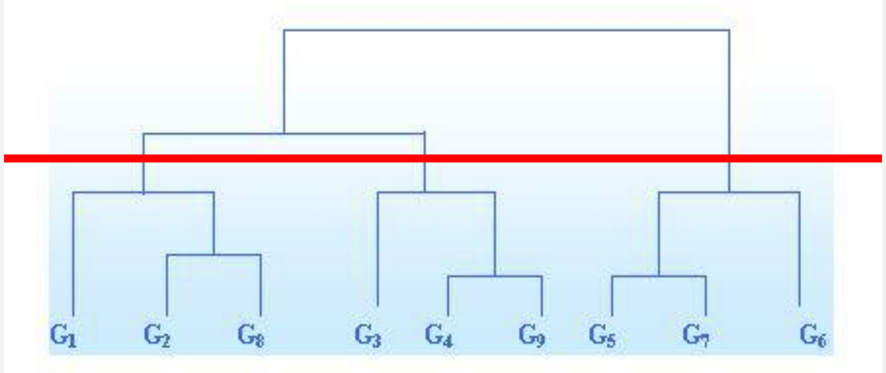


Modified method with minimum-distance measure

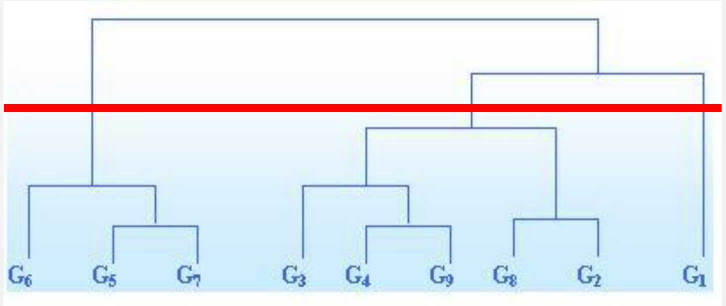


Modified method with maximum-distance measure

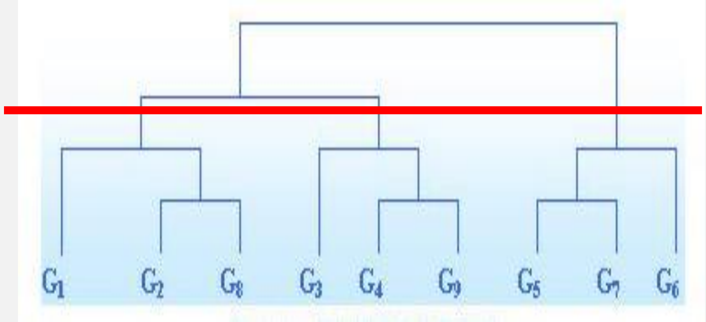
Comparison of the results from AGNES and its modified method



Original method



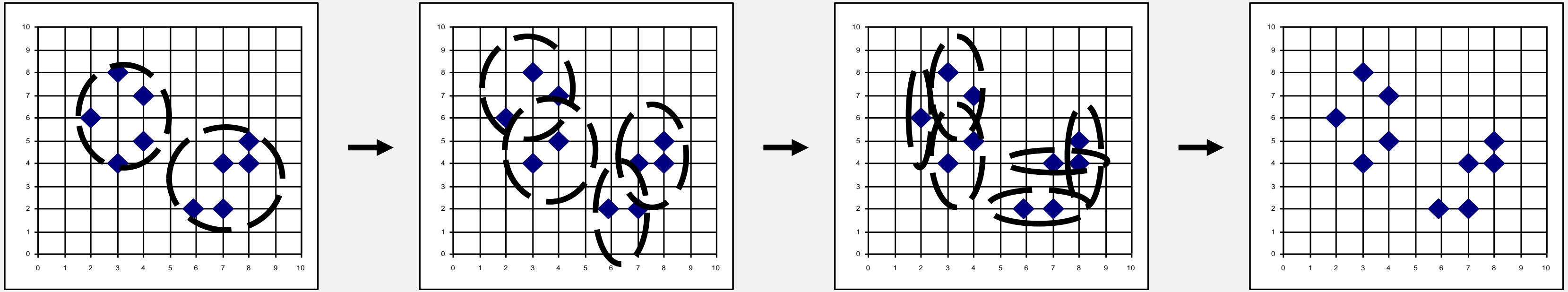
Modified method with
minimum-distance measure



Modified method with
maximum-distance measure

DIANA (DIvisive ANAlysis)

- Implemented in statistical analysis packages, e.g., SPSS, Splus
- Inverse order of AGNES
- Eventually each node forms a cluster on its own



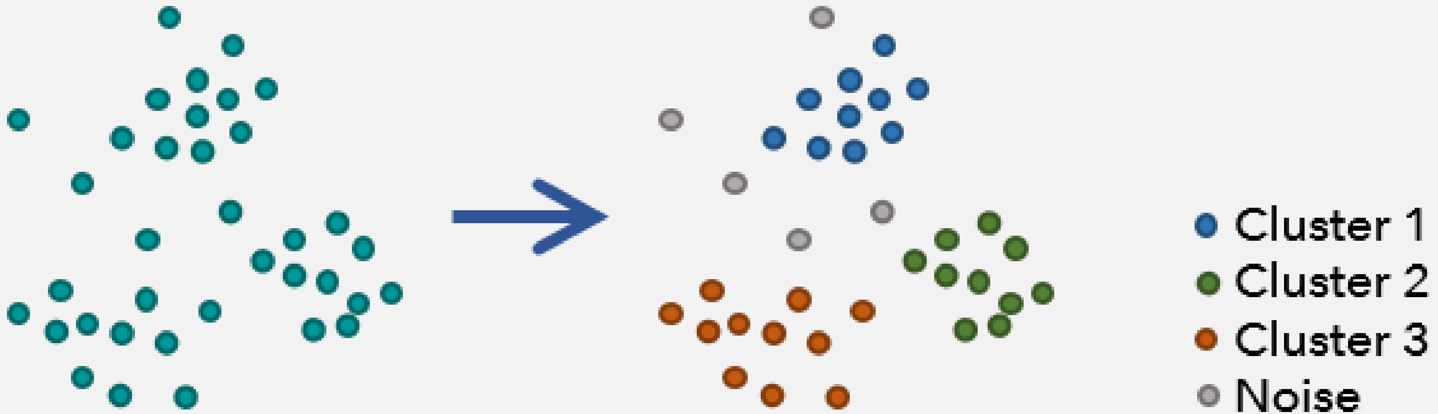
Both AGNES and DIANA methods can cluster objects and variables.

Lect. 8 Spatial Clustering Analysis

- What is Clustering Analysis?
- How to Measure the Similarity of Clustering
- Spatial Clustering Approaches
  - Partitioning Clustering
  - Hierarchical Clustering
  - Density-based Clustering
  - Spatially Constrained Multivariate Clustering
- Cases

Density-based Clustering

- Finds clusters of point features within surrounding noise based on their spatial distribution.
- Time can also be incorporated to find space-time clusters.



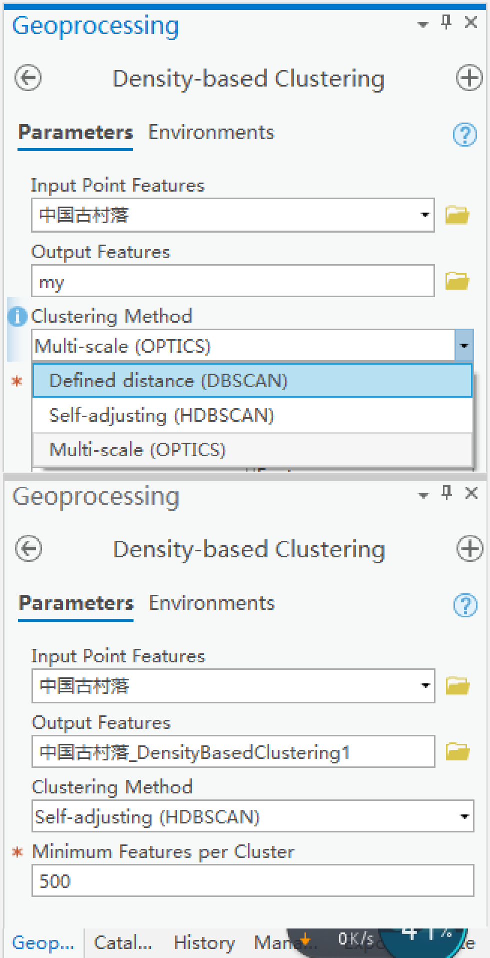
Density-based Clustering

- The Density-based Clustering tool works by detecting areas where points are concentrated and where they are separated by areas that are empty or sparse. Points that are not part of a cluster are labeled as noise.
- This tool uses unsupervised machine learning clustering algorithms which automatically detect patterns based purely on spatial location and the distance to a specified number of neighbors.
- These algorithms are considered unsupervised because they do not require any training on what it means to be a cluster.

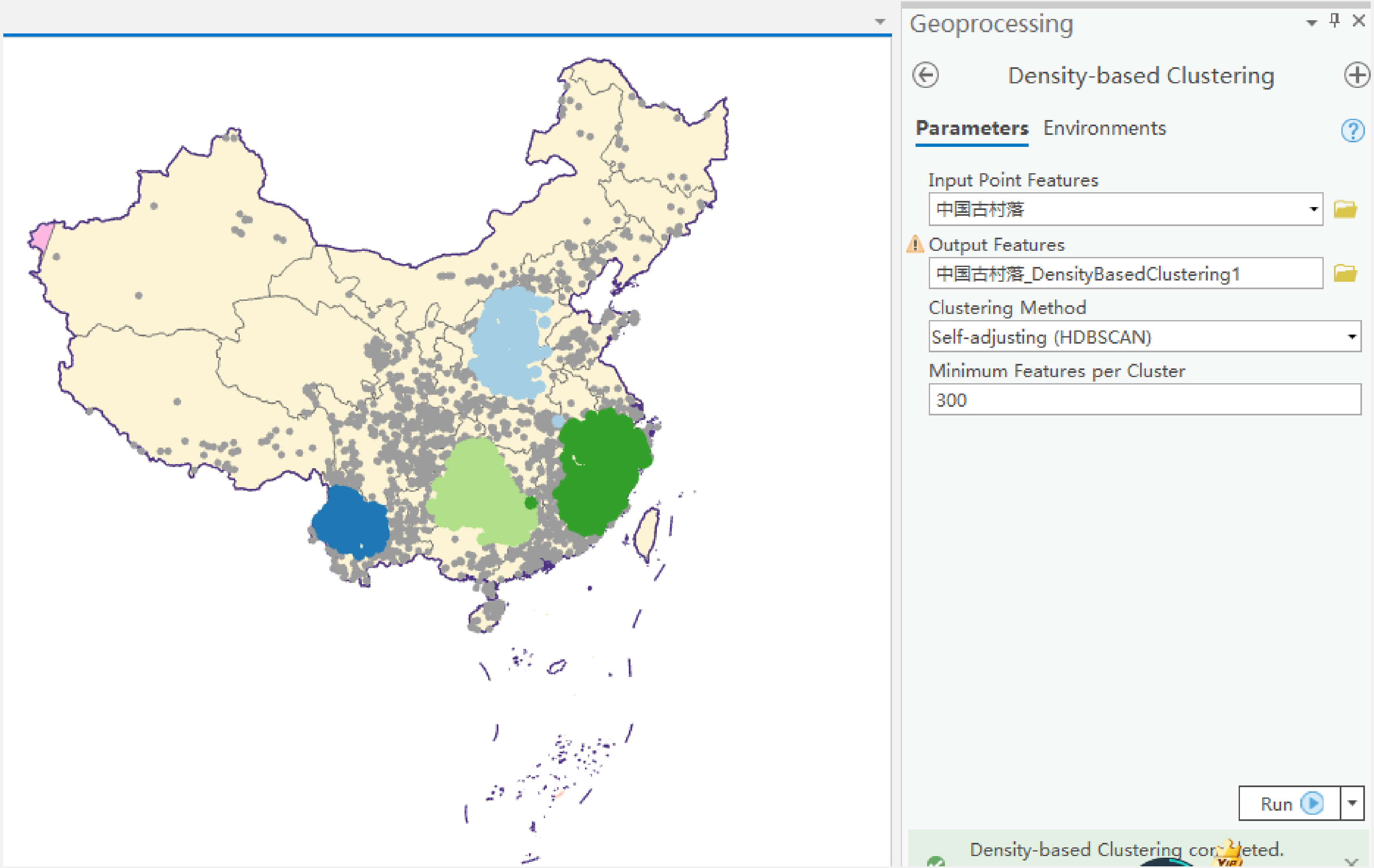
Potential Applications of Density-based Clustering

- Urban water supply networks are an important hidden underground asset. The clustering of pipe ruptures and bursting can indicate looming problems. Using the Density-based Clustering tool, an engineer can find where these clusters are and take pre-emptive action on high-danger zones within water supply networks.
- Geo-locating tweets following natural hazards or terror attacks can be clustered and rescue and evacuation needs can be informed based on the size and location of the clusters identified.
- Say you are studying a particular pest-borne disease and have a point dataset representing households in your study area, some of which are infested, some of which are not. By using the Density-based Clustering tool, you can determine the largest clusters of infested households to help pinpoint an area to begin treatment and extermination of pests.

Density-based Clustering



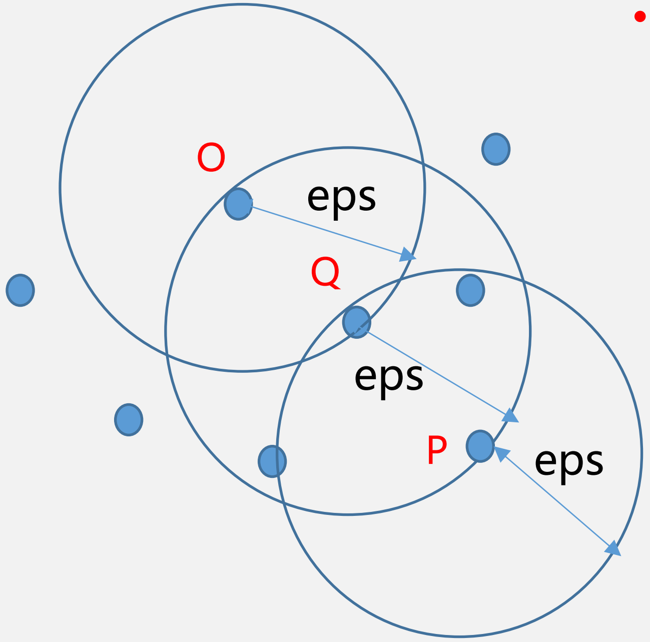
- **Defined distance (DBSCAN)-** Density-Based Spatial Clustering of Application with Nosie)
  - Uses a specified distance to separate dense clusters from sparser noise.
  - DBSCAN is the fastest of the clustering methods but is only appropriate if there is a very clear distance to use that works well to define all clusters that may be present. This results in clusters that have similar densities.
  - DBSCAN also allows us to use the **Time Field** and **Search Time Interval** parameters to find clusters of points in space and time.
- **Self-adjusting (HDBSCAN)-**Hierarchical Density-Based Spatial Clustering of Applications with Noise
  - Uses varying distances to separate clusters of varying densities from sparser noise.
  - HDBSCAN is the most data-driven of the clustering methods and requires the least user input.
- **Multi-scale (OPTICS)-** Ordering Points To Identify the Clustering Structure
  - Uses the distance between neighbors and a reachability plot to separate clusters of varying densities from noise.
  - OPTICS offers the most flexibility in fine-tuning the clusters that are detected, though it is computationally intensive, particularly with a large Search Distance.



Geoprocessing
Density-based Clustering
Parameters
Environments
Input Point Features
中国古村落
Output Features
中国古村落_DensityBasedClustering1
Clustering Method
Self-adjusting (HDBSCAN)
Minimum Features per Cluster
300
Run

Some Concepts for DBSCAN

- **e-neighborhood:** The e-neighborhood of any point in the space is the region contained by a circle with that point as the center of the circle and the **eps** as the radius.

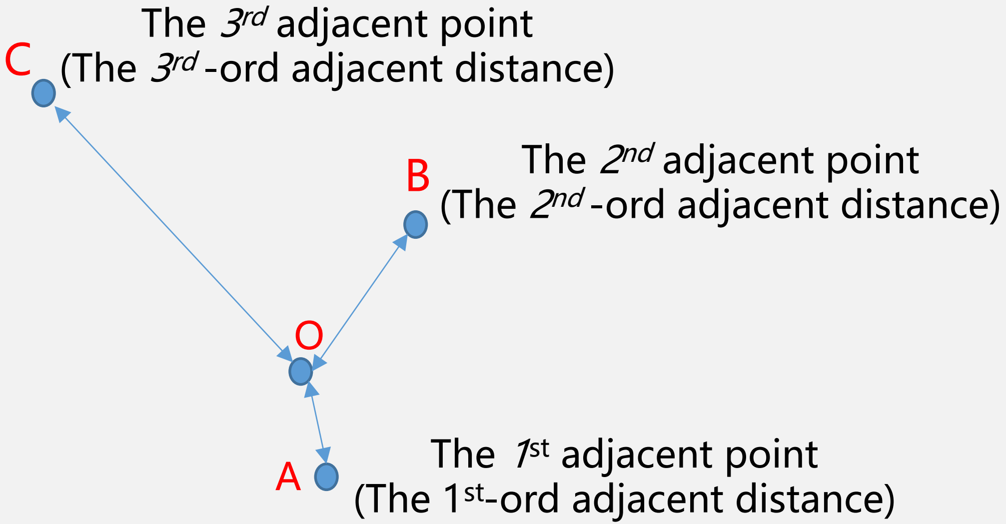


- **Point Density :** The point density is the ratio of the number of points contained in the e-neighborhood to the area of that neighborhood. The number of points contained in the e-neighborhood can be regarded as point density.

- **Core-point:** A point is called a core point if the number of points within its eps is greater than threshold (MinPts).
  - Suppose MinPts =3, Q and P are core points, while O is non-core point.

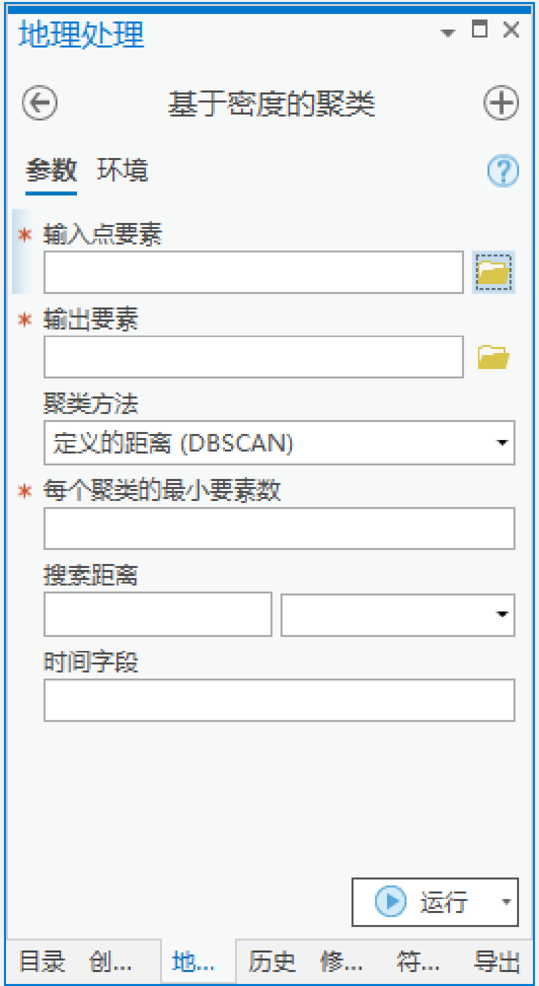
Some Concepts for DBSCAN

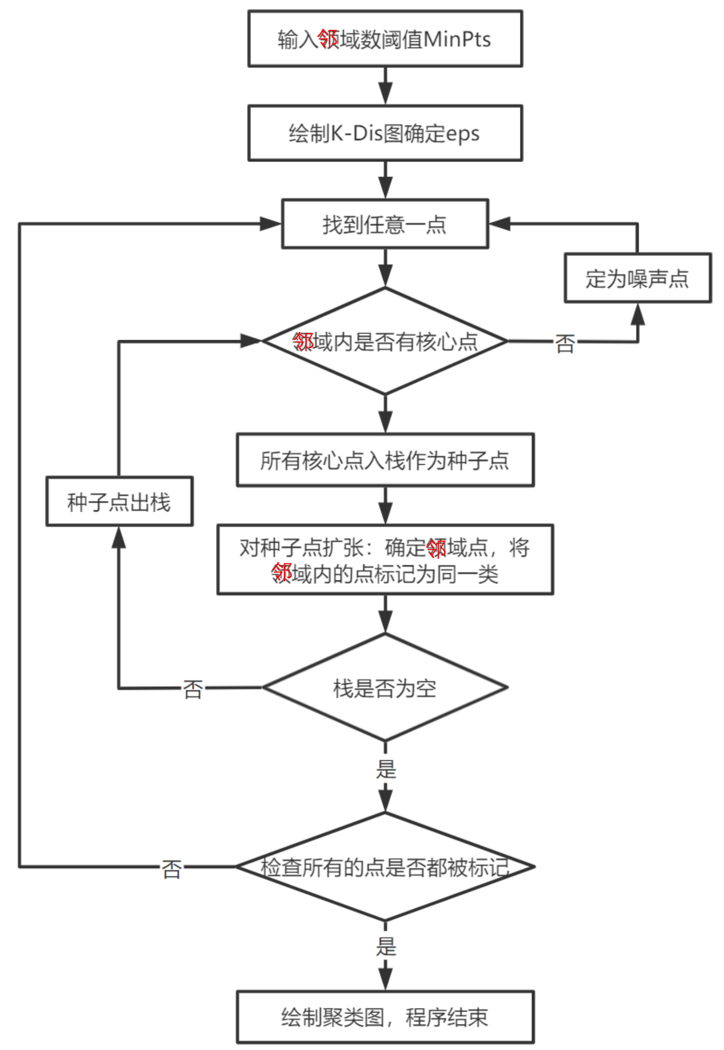
- ***K**th***-order neighbor distance:** The distance between a point and its *k*th adjacent point.



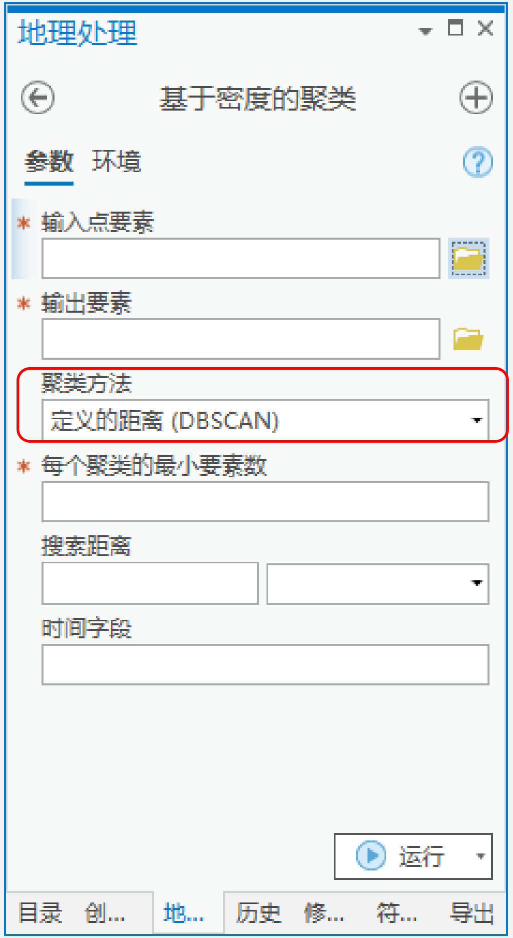
- ***K*-Dis plot:** It is a schematic plot of sorting the *k*th-order **neighbor** distances of each point.

The procedure of DBSCAN





The Parameter of Minimum Features per Cluster for DBSCAN



- This parameter determines the minimum number of features required to consider a grouping of points a cluster.
  - For instance, if you have a number of different clusters, ranging in size from 10 points to 100 points, and you choose a Minimum Features per Cluster of 20, all clusters that have less than 20 points will either be considered noise (because they do not form a grouping large enough to be considered a cluster) or they will be merged with nearby clusters in order to satisfy the minimum number of features required.
  - In contrast, choosing a Minimum Features per Cluster smaller than what you consider your smallest meaningful cluster may lead to meaningful clusters being divided into smaller clusters. In other words, the smaller the Minimum Features per Cluster, the more clusters will be detected.
- The larger the Minimum Features per Cluster, the fewer clusters will be detected.

The Parameter of Minimum Features per Cluster for Calculating Core-Distance of DBSCAN

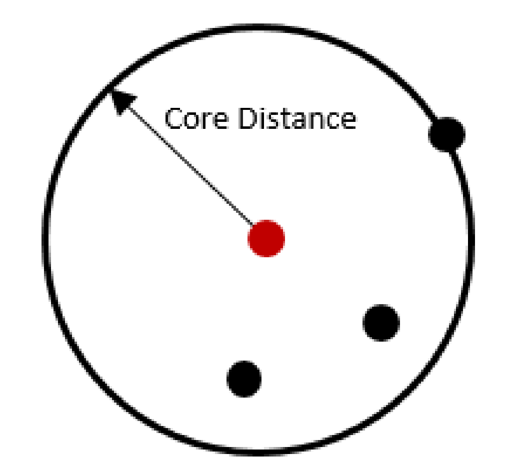
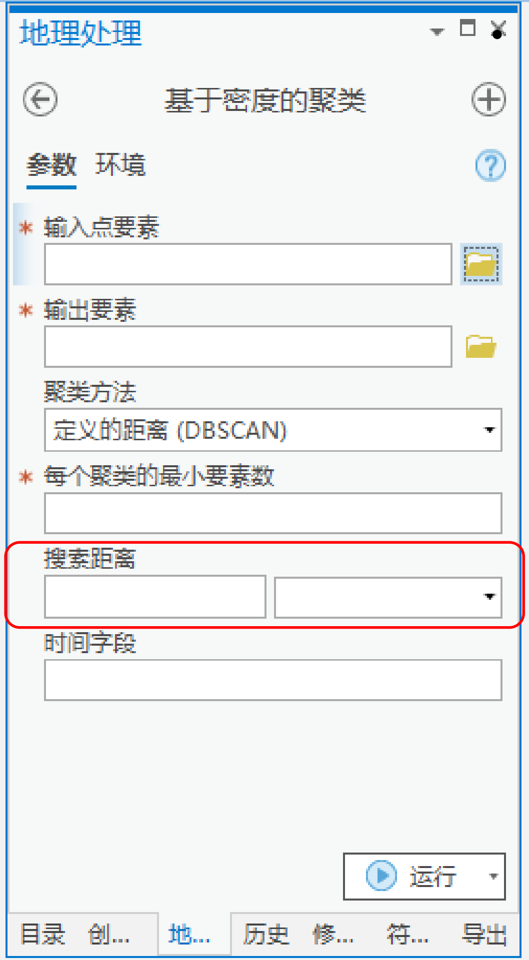


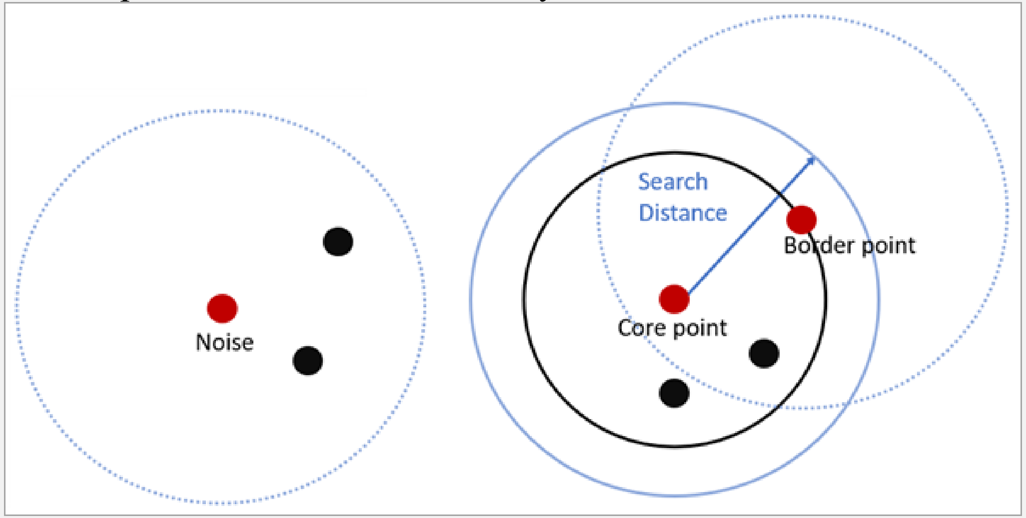
Illustration of the core-distance, measured as the distance from a particular feature that must be traveled to create a cluster with a minimum of 4 features including itself.

- The **Minimum Features per Cluster** parameter is also important in the calculation of the core-distance, which is a measurement used by all three methods to find clusters.
- Conceptually, the core-distance for each point is a measurement of the distance that is required to travel from each point to the defined minimum number of features.
- So, if a large **Minimum Features per Cluster** is chosen, then the corresponding core-distance will be larger.
- If a small **Minimum Features per Cluster** is chosen, then the corresponding core-distance will be smaller.
- The core-distance is related to the **Search Distance** parameter, which is used by both the **Defined distance (DBSCAN)** and **Multi-scale (OPTICS)** methods.

Search Distance for DBSCAN



For **DBSCAN**, if the **Minimum Features per Cluster** can be found within the **Search Distance** from a particular point, that point will be marked as a core-point and included in a cluster, along with all points within the core-distance. A border-point is a point that is within the search distance of a core-point but does not itself have the minimum number of features within the search distance. Each resulting cluster is composed of core-points and border-points, where core-points tend to fall in the middle of the cluster and border-points fall on the exterior. If a point does not have the minimum number of features within the search distance and is not a within the search distance of another core-point (in other words, it is neither a core-point nor a border-point), it is marked as a noise point and not included in any cluster.



Lect. 8 Spatial Clustering Analysis

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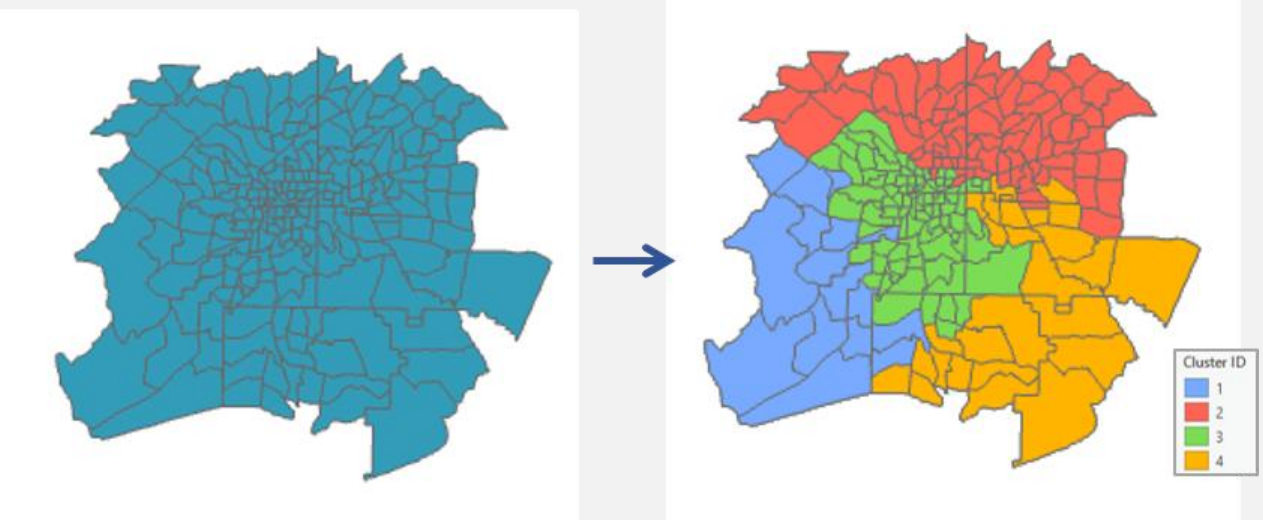
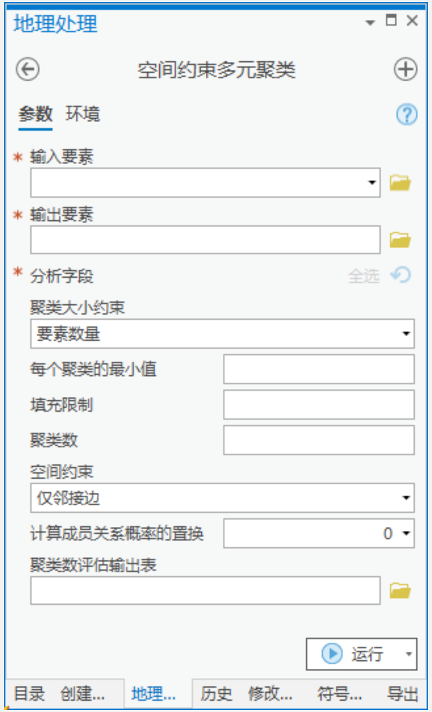
- □ Cases

Spatially Constrained Multivariate Clustering

Finds spatially contiguous clusters of features based on a set of feature attribute values and optional cluster size limits.





Spatially Constrained Multivariate Clustering

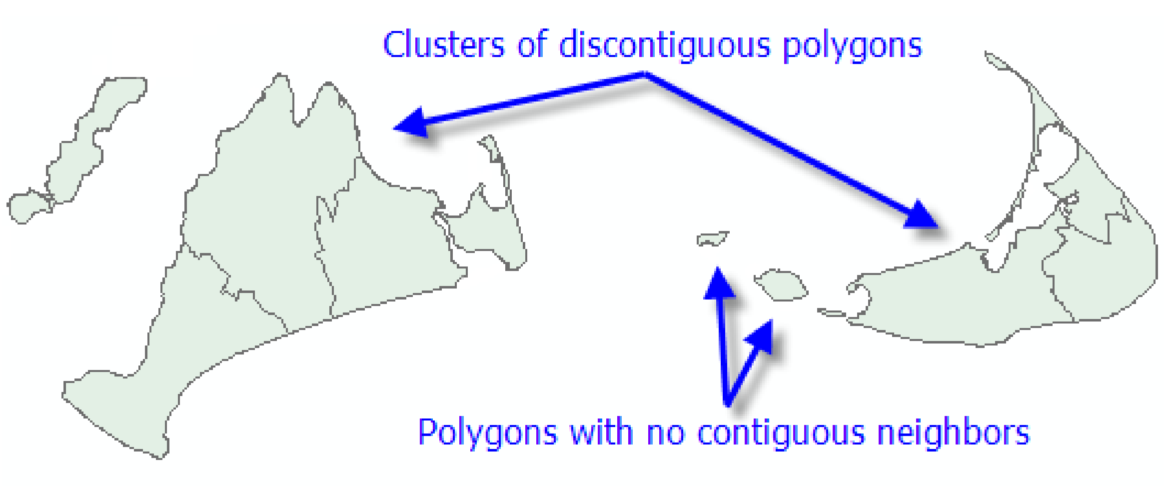
- Given the number of clusters to create, it will look for a solution where all the features within each cluster are as similar as possible, and all the clusters themselves are as different as possible.
- Feature similarity is based on the set of attributes specified for the **Analysis Fields** parameter and may optionally incorporate constraints on the size of the clusters.
- The algorithm employs a connectivity graph (minimum spanning tree) and a method called SKATER to find natural clusters in your data as well as evidence accumulation to evaluate cluster membership likelihood.

Spatially Constrained Multivariate Clustering

- Clustering, grouping, and classification techniques are some of the most widely used methods in machine learning.
- The Spatially Constrained Multivariate Clustering tool uses unsupervised machine learning methods to determine natural clustering in your data.
- These classification methods are considered unsupervised, as they do not require a set of preclassified features to guide or train on in order to determine the clustering of your data.

Spatially Constrained Multivariate Clustering

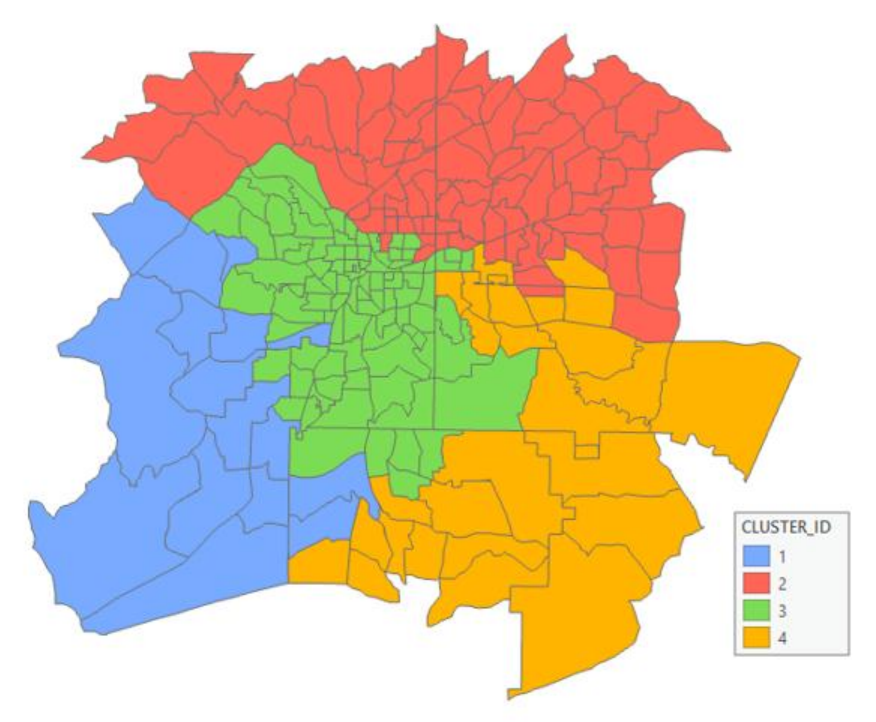
- The **Spatial Constraints** parameter ensures that the resultant clusters will be spatially proximal.
- The **Contiguity** options are enabled for polygon feature classes and indicate that features can only be part of the same cluster if they share an edge (**Contiguity edges only**) or if they share either an edge or a vertex (**Contiguity edges corners**) with another member of the cluster.
- The polygon contiguity options are not good choices, however, if your dataset includes clusters of discontiguous polygons or polygons with no contiguous neighbors at all.



Spatial Constraints for Constrained Multivariate Clustering

- **Contiguity edges only** —Clusters will contain contiguous polygon features. Only polygons that share an edge can be part of the same cluster.
- **Contiguity edges corners**— Clusters will contain contiguous polygon features. Only polygons that share an edge or a vertex can be part of the same cluster. This is the default for polygon features.
- **Trimmed Delaunay triangulation**— Features in the same cluster will have at least one natural neighbor in common with another feature in the cluster. Natural neighbor relationships are based on a trimmed Delaunay triangulation. Conceptually, Delaunay triangulation creates a nonoverlapping mesh of triangles from feature centroids. Each feature is a triangle node, and nodes that share edges are considered neighbors. These triangles are then clipped to a convex hull to ensure that features cannot be neighbors with any features outside the convex hull. This is the default for point features.
- **Get spatial weights from file**— Spatial, and optionally temporal, relationships are defined by a specified spatial weights file (.swm). Create the spatial weights matrix using the Generate Spatial Weights Matrix or Generate Network Spatial Weights tool. The path to the spatial weights file is specified by the Spatial Weights Matrix File parameter.

The Results of Constrained Multivariate Clustering



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- ☐ Cases

Cases

- Spatiotemporal analysis of activity-travel fragmentation based on spatial clustering and sequence analysis Spatial clustering-2022
- Combining spatial clustering and spatial regression models to understand distributional inequities in access to urban green spaces-2025
- A machine learning method for the prediction of ship motion trajectories in real operational conditions-2023
- Decoding spatial patterns of urban thermal comfort: Explainable machine learning reveals drivers of thermal perception-2025

References

1. Chapter 10 in Rogerson’ s 《Statistical methods for geography》.
2. ArcGIS Pro Help
3. Spatiotemporal analysis of activity-travel fragmentation based on spatial clustering and sequence analysis Spatial clustering-2022
4. Combining spatial clustering and spatial regression models to understand distributional inequities in access to urban green spaces-2025
5. A machine learning method for the prediction of ship motion trajectories in real operational conditions-2023
6. Decoding spatial patterns of urban thermal comfort: Explainable machine learning reveals drivers of thermal perception-2025

…

Reading, Exercises &Lab

1. What is the basic idea of cluster analysis？ How to measure the distance among clusters？ What are the applications of cluster analysis?
2. Try to explain at least 3 clustering methods (e.g., *k*-means clustering, density-based clustering and others).
3. What is the difference between clustering and classification?
4. Use maximum distance to cluster 6 objects list in the table into 2 categories by modified AGNES method.
5. Do Lab 8.

	O1	O2	O3	O4	O5	O6
O1	0					
O2	0.375	0				
O3	0.483	0.776	0			
O4	1.749	1.596	1.926	0		
O5	1.516	1.336	1.662	0501	0	
O6	1.972	1.743	2.154	0.693	0.589	0